



Image Caption Generation Model

Clip+GPT-2

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Our goal: generate captions from images

Combine 2 powerful model: CLIP and GPT-2

We use multiple ML algorithms from class and beyond:

- Gradient Descent
- Maximum Likelihood Estimation

We are going to introduce our model structure, loss function and show our progress by demo!

Image Caption Generator

Upload an image and get an AI-generated caption



Generated Caption:

A road with a view of mountains and a road sign.

The Problem of Unlabeled Images



1. On modern digital platforms like AirDrop, images are constantly shared.
2. These images usually lack simple filenames — often just numbers or codes.
3. As these unlabeled images pile up, managing them becomes time-consuming
4. This reduces productivity and makes information search difficult for individuals and organizations.

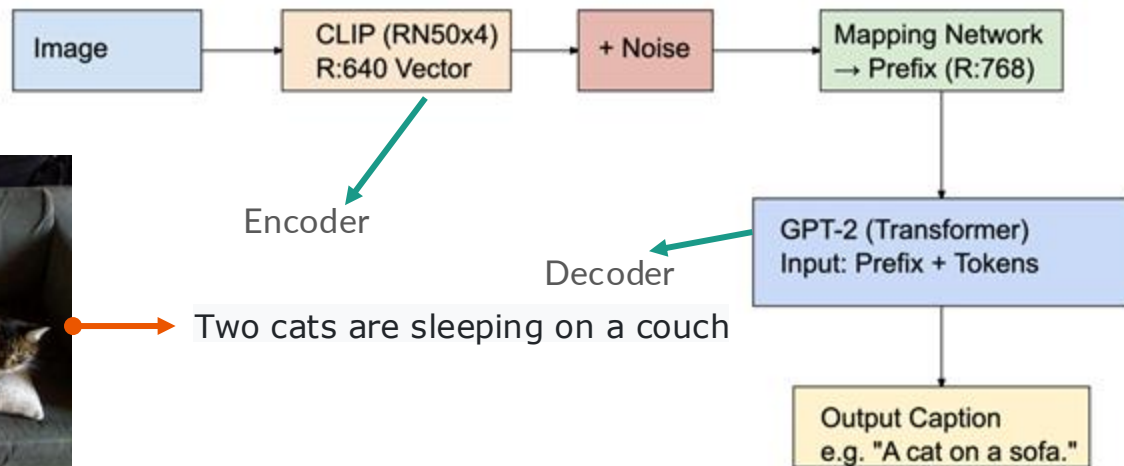
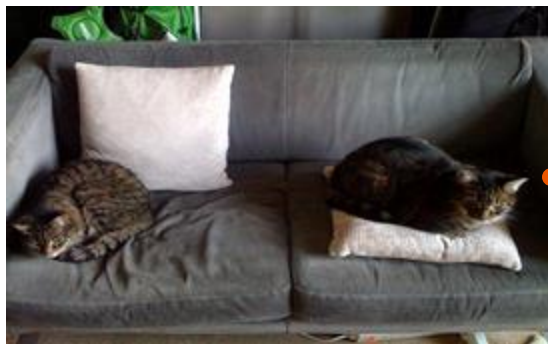
Core Principle

1. Bridging Visual and Linguistic Spaces with Noisy Mapping

More robust and reduce overfitting

The Equation:

Better Captions = CLIP F

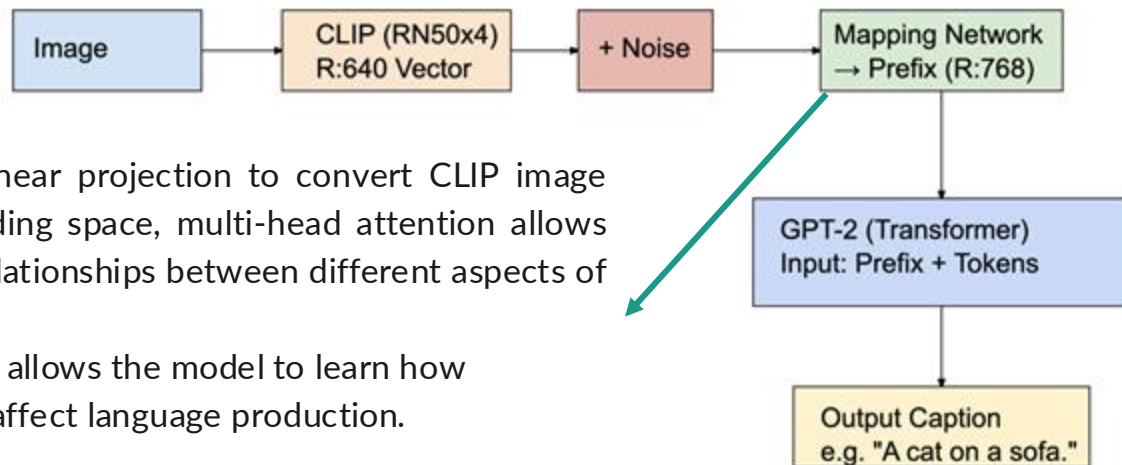


Core Principle

2. Multi-Head Attention as a Cross-Modal Connector

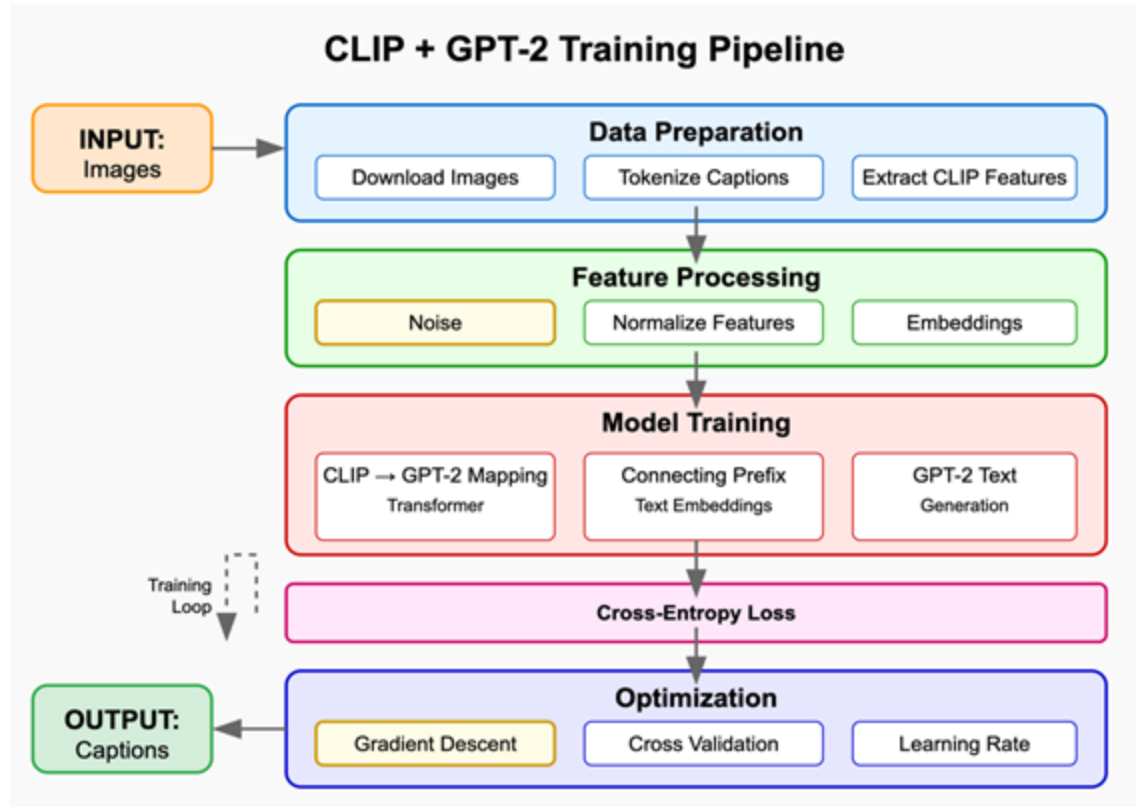
The Equation:

Better Captions = CLIP F



- Rather than using a simple linear projection to convert CLIP image features into GPT-2's embedding space, multi-head attention allows our model to learn complex relationships between different aspects of the visual representation.
- Multi-head Attention Network allows the model to learn how different regions in the image affect language production.

Our Project Pipeline



Loss and performance evaluation

Training: cross-entropy loss

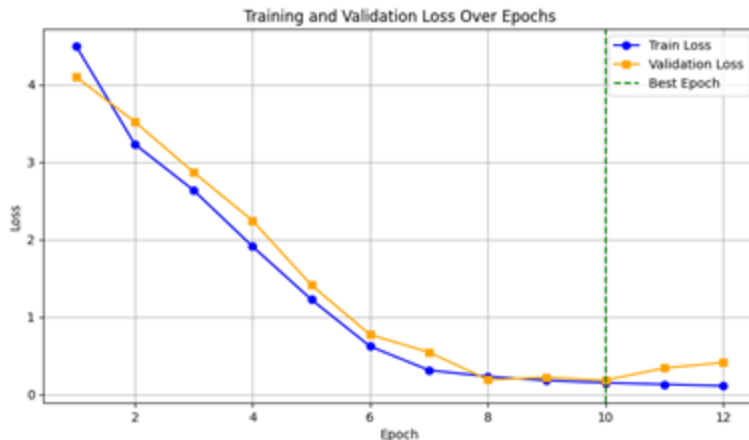
Goal: Maximize the conditional probability that each correct token occurs

$$\text{Loss} = - \sum_{t=1}^T \log P_{\text{GPT-2}}(w_t \mid w_{<t}, \text{prefix})$$

Predict the probability of the t-th token conditional on the prefix and the history of the token before

$$\hat{\theta} = \arg \min_{\theta} - \sum_{i=1}^N \sum_{t=1}^{T_i} \log P_{\theta}(w_t^{(i)} \mid w_{<t}^{(i)}, \text{prefix}^{(i)})$$

(Maximum Likelihood Estimate)



Cross Entropy
Loss

Validation Loss

Best model
output

Loss and performance evaluation

Predicting: BLEU + CIDEr

	What It Measures	Typical Range
BLEU-1	single word overlap between generated caption and reference	0-1
BLEU-2	two-word sequence overlap	0-1
BLEU-3	three-word sequence overlap	0-1
BLEU-4	Four-word sequence overlap	0-1
CIDEr	Consensus-based evaluation using TF-IDF weighting	0-2

- BLEU-n: Higher means that the caption is closer to the reference, above 0.6 is better.
- CIDEr: >1 indicates that the generated content is highly consistent with the reference; >1.5 very good

Prediction scores:

Model Parameter:

Transformer Layers: 8

Learning rate: $5e-5$

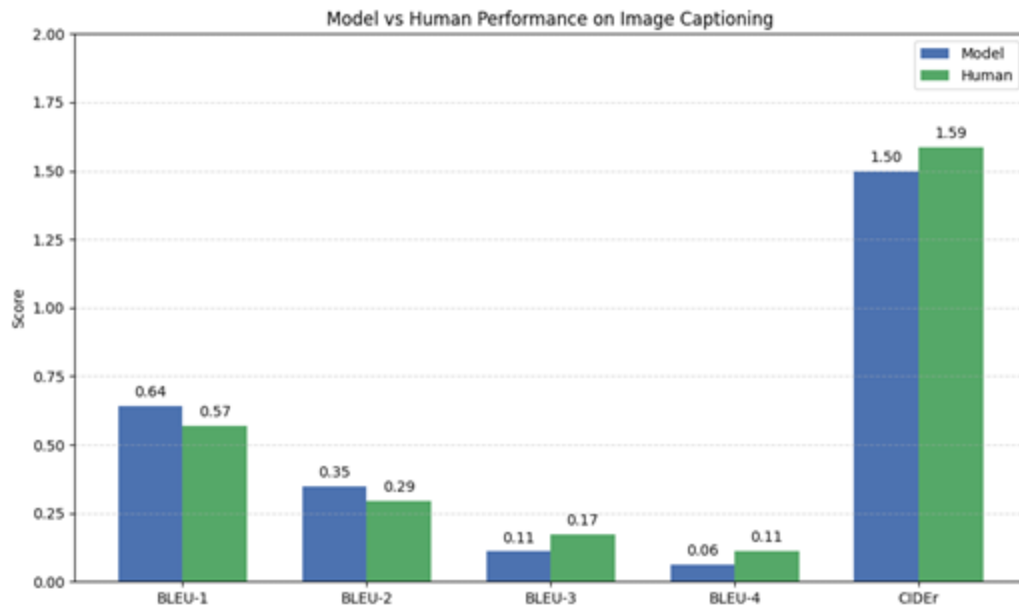
Prefix length: 40

Noise rate: 10%

Dataset: COCO Image

Training set: 80, 000 images with 5 captions per image.

Validation set: 2,000 images



- Human captions show better performance on longer word sequences, indicating more natural grammatical structure.
- The CIDEr score shows our model can achieve 94.4% of human-level consensus quality.



Demo

Why Our Findings Matter



1. Automated organization of visual content:

Our model efficiently manages unlabeled images, reducing the need for manual effort.

2. Effective CLIP-to-GPT-2 Mapping :

Our multi-head attention mechanism bridges visual and textual modalities for high-quality captions.

3. Noise Injection Boosts Generalization :

Introducing noise during training boosts generalization and helps create diverse, accurate captions.



Thank you